

Session 3 Solutions

Lecture 31

1. *How are encryption and decryption performed with a Feistel cipher? Prove that decryption undoes encryption.*

A Feistel cipher starts with an initial permutation and then performs a number of encryption steps with a sub-key which is generated from the encryption key. It then performs the reverse permutation. Each step splits the data into left and right halves. The new left half is the old left half xored with a function of the right half and the sub-key, while the right half remains unchanged. The two halves are then swapped. Decryption performs the same steps but with the sub-keys in reverse order.

The initial permutation undoes the reverse of the initial permutation. Swapping twice does not change the data. Xor with the same function twice has no effect. The keys are used in reverse order so that the last encrypted step is undone by the first decrypted step because they are both using the same key. This applies to all the steps.

2. *Describe briefly how the DES algorithm implements the Feistel scheme, mentioning, ~~input~~, the role of the key length and also the so-called S-Boxes. You do not need to provide details of the actual S-Boxes, rather describe in general terms their role in the algorithm.*

DES has 16 complex steps, alternating with simple permutations. Separate sub-keys are used for each of the complex steps. It has a key size of 56 bits, reduced from the original 128 bits after advice from the USA NSA. The S-Boxes are just look-up tables, designed for good confusion and diffusion.

3. *Imagine you have the task of designing a hardware encryption device based on DES chips. Your company anticipates that a well-funded organisation will make a serious attempt to break the data encrypted by your device. How would you choose to employ the DES chips?*

DES itself is rather weak because of the short key. It can be modified to use a longer key and so a combination method such as two-key triple DES or three-key triple DES could be used.

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4. *Describe how the process of choosing the AES algorithm was different from the one that led to DES. Was the new process better or worse? In what ways is AES better than DES?*

DES was secretly commissioned, while the process that led to AES was public. There is thus more confidence that the AES process is more secure, even though no backdoors in DES have been discovered. The main problem with DES was the short key length, which has been fixed with AES.

5. *Describe how the Rijndael AES encryption algorithm uses polynomial arithmetic. What is the advantage of using this arithmetic over the use of S-boxes in DES?*

8-bit polynomial arithmetic is used to transform single data bytes in the SubBytes stage. Polynomials of degree 4 where the coefficients are 8-bit polynomials are used in the MixColumns stage. These polynomials are reduced mod $X^4 + 1$, which is not irreducible.